Question	Answer	Marks	Partial Marks
20	$x^2+6x-40$ [=0] or $y^2-40y-41$ [=0]	**M2**	**M1** for correct method to eliminate one variable e.g. $x^2-2(11-3x)=18$ or $\frac{(11-y)^2}{3^2}-2y=18$
	$(x-4)(x+10)$ [=0] or $(y-41)(y+1)$ [=0]	**M1**	or for correct factors for *their* quadratic equation or for correct use of quadratic formula for *their* quadratic equation or for correctly completing the square for *their* quadratic equation
	$x = 4, y = -1$ $x = -10, y = 41$	**B2**	**B1** for $x = 4, x = -10$ or for $y = -1, y = 41$ or for a correct pair of x and y values If B0 scored and at least 1 method mark scored **SC1** for correct substitution shown of both of *their x* values or *their y* values into $3x + y = 11$ or $x^2 - 2y = 18$
21(a)	35.1 or 35.05 to 35.06	**4**	**M3** for $\tan = \frac{14.5}{\sqrt{18.6^2+9^2}}$ oe or **M2** for $\left[AC^2=\right]18.6^2+9^2$ oe or better or $\left[AG^2=\right]18.6^2+9^2+14.5^2$ or **M1** for recognising the angle *GAC*

Question	Answer	Marks	Partial Marks
21(b)	$30-\sqrt{18.6^2+9^2+14.5^2}$ $30-\dfrac{14.5}{\sin(their\,\mathbf{(a)})}$ or $30-\dfrac{\sqrt{18.6^2+9^2}}{\cos(their\,\mathbf{(a)})}$	**M2**	**M1** for $AG^2=18.6^2+9^2+14.5^2$ oe or better or $\sin(their\,\mathbf{(a)})=\dfrac{14.5}{\text{AG}}$ or $\cos(their\,\mathbf{(a)})=\dfrac{\sqrt{18.6^2+9^2}}{\text{AG}}$
	4.75 to 4.78…	**A1**	
22(a)	$\boldsymbol{a}-\dfrac{2}{5}\boldsymbol{b}$ oe simplified	**2**	**M1** for $-\boldsymbol{b}+\boldsymbol{a}+\dfrac{3}{5}\boldsymbol{b}$ or a correct route
22(b)	$\dfrac{5}{2}\boldsymbol{a}$ oe	**2**	**B1** for k**a** where $k>1$ or $\dfrac{5}{2}\overrightarrow{\text{OR}}$